Paper 26 — Normalization and Continuum Limit of the Entanglement Laplacian

From Mutual-Information Graphs to the Laplace–Beltrami Operator

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Abstract

Bottom-Up Quantum Gravity (BuP) starts from an entanglement graph $W_{ij} = I(i : j)$ and defines an entanglement Laplacian

$$L_{\text{ent}} = D - W, \qquad D_{ii} = \sum_j W_{ij}.$$

Paper 25 identified the continuum limit

$$c_{N,\epsilon} L_{N,\epsilon} \longrightarrow -\Delta_g$$

as one of the central mathematical debts of the programme. This paper isolates and tests the normalization $c_{N,\epsilon}$ for local Gaussian graph Laplacians. For a compact D-dimensional Riemannian manifold with locally uniform node density $\rho = N/\mathrm{Vol}(M)$, we derive the continuum expansion

$$L_{N,\epsilon} f \simeq -\rho(4\pi)^{D/2}\epsilon^{D/2+1}\Delta_g f,$$

hence

$$\boxed{c_{N,\epsilon} = \frac{1}{\rho(4\pi)^{D/2}\epsilon^{D/2+1}}.}$$

The formula is validated on three controlled geometries: S^1, T^2, and S^2. We then test a deliberately non-uniform S^1 density and show that the raw combinatorial Laplacian no longer converges to a pure Laplace–Beltrami operator, but to a density-biased diffusion operator with a drift term proportional to $\nabla \log \rho \cdot \nabla$. Finally, we show that a symmetric density correction $W_{ij}/\sqrt{q_i q_j}$ reduces this bias in a controlled non-uniform test. The paper therefore establishes the precise normalization regime required for the BuP entanglement Laplacian and clarifies the density-correction needed before applying the result to non-uniform astrophysical or lensing graphs.

Contents

1 Introduction

The Bottom-Up Quantum Gravity programme is built around the idea that geometry is not fundamental but emerges from quantum-information structure. The microscopic object is the mutual-information graph

$$W_{ij} = I(i:j), \tag{1}$$

from which one defines an entanglement Laplacian

$$L_{\text{ent}} = D - W, \qquad D_{ii} = \sum_j W_{ij}. \tag{2}$$

Paper 25 reformulated the programme as a variational theory of entanglement equilibrium. One of its central continuum requirements is the convergence

$$c_{N,\epsilon} L_{N,\epsilon} \to -\Delta_g, \tag{3}$$

where Δ_g is the Laplace–Beltrami operator of the emergent geometry.

The purpose of Paper 26 is to isolate this claim from the rest of the gravitational programme and test it under controlled mathematical assumptions. We do not claim here that arbitrary mutual-information graphs automatically converge to Laplace–Beltrami operators. Instead, we prove and test the normalization for local Gaussian kernel graphs, then identify the additional hypotheses required for BuP graphs.

2 Role in the BuP programme

The convergence of the entanglement Laplacian controls the chain

$$W_{ij} \longrightarrow L_{\text{ent}} \longrightarrow -\Delta_g \longrightarrow (-\Delta_g)^{-1} \longrightarrow \Phi_{\text{BuP}}. \tag{4}$$

This chain supports several later components of the programme:

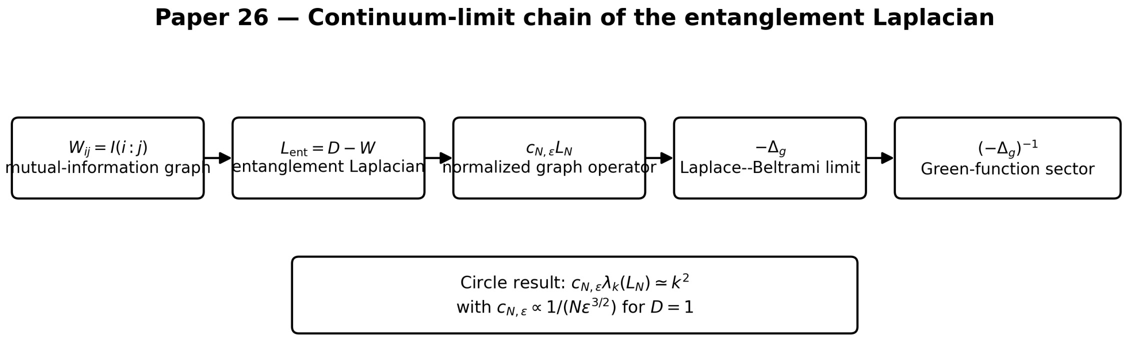

Figure 1: Continuum-limit chain tested in Paper 26. The goal is to normalize the entanglement graph Laplacian so that it converges to the Laplace–Beltrami operator, thereby supporting the Green-function sector used in the weak-field BuP programme.

- the weak-field Green-function construction;
- the effective propagator law;
- the galactic and lensing sectors;
- the correction tensor $\mathcal{H}^{\rm BuP}_{\mu\nu}$;
- the effective Einstein-limit target of Paper 25.

Thus Paper 26 addresses a narrow but crucial question:

> *Under what normalization does a local entanglement graph Laplacian become the Laplace–Beltrami operator?*

3 Kernel graph setup

Let M be a compact D-dimensional Riemannian manifold with metric g. Let $x_i \in M$, $i = 1, \ldots, N$, be sampled with node density ρ. We first consider the locally uniform case

$$\rho = \frac{N}{\mathrm{Vol}(M)}. \tag{5}$$

Define the local Gaussian kernel

$$W_{ij}^{(\epsilon)} = \exp\left[-\frac{d_g(x_i, x_j)^2}{4\epsilon}\right], \tag{6}$$

with the diagonal removed,

$$W_{ii} = 0. \tag{7}$$

The unnormalized graph Laplacian is

$$(L_{N,\epsilon} f)_i = \sum_j W_{ij}^{(\epsilon)} (f_i - f_j). \tag{8}$$

This sign convention makes $L_{N,\epsilon}$ positive semidefinite and its continuum limit the positive operator $-\Delta_g$.

4 Continuum expansion

In local normal coordinates around x, set $u = y - x$. For a smooth test function f,

$$f(x+u) = f(x) + u^a \partial_a f(x) + \frac{1}{2} u^a u^b \partial_a \partial_b f(x) + O(|u|^3). \tag{9}$$

The graph Laplacian is approximated by

$$L_{N,\epsilon} f(x) \simeq \rho \int_{\mathbb{R}^D} e^{-|u|^2/(4\epsilon)} \left[f(x) - f(x+u) \right] \mathrm{d}^D u. \tag{10}$$

The first-order term vanishes by symmetry,

$$\int_{\mathbb{R}^D} u^a e^{-|u|^2/(4\epsilon)} \mathrm{d}^D u = 0. \tag{11}$$

The second-order Gaussian moment is

$$\int_{\mathbb{R}^D} u^a u^b e^{-|u|^2/(4\epsilon)} \mathrm{d}^D u = 2\epsilon (4\pi\epsilon)^{D/2} \delta^{ab}. \tag{12}$$

Therefore

$$\begin{aligned} L_{N,\epsilon} f(x) &\simeq -\rho \frac{1}{2} \partial_a \partial_b f(x) \int_{\mathbb{R}^D} u^a u^b e^{-|u|^2/(4\epsilon)} \mathrm{d}^D u, & (13) \\ &= -\rho\epsilon (4\pi\epsilon)^{D/2} \Delta_g f(x). & (14) \end{aligned}$$

Thus

$$\boxed{L_{N,\epsilon} f \simeq -\rho (4\pi)^{D/2} \epsilon^{D/2+1} \Delta_g f.} \tag{15}$$

5 General normalization law

Equation (15) implies the normalization

$$\boxed{c_{N,\epsilon} = \frac{1}{\rho (4\pi)^{D/2} \epsilon^{D/2+1}}, \qquad \rho = \frac{N}{\mathrm{Vol}(M)}.} \tag{16}$$

Equivalently,

$$c_{N,\epsilon} = \frac{\mathrm{Vol}(M)}{N (4\pi)^{D/2} \epsilon^{D/2+1}}. \tag{17}$$

With this convention,

$$\boxed{c_{N,\epsilon} L_{N,\epsilon} f \to -\Delta_g f} \tag{18}$$

in the locally uniform density regime.

6 Validation on S^1

For the unit circle,

$$\mathrm{Vol}(S^1) = 2\pi, \qquad D = 1, \qquad \rho = \frac{N}{2\pi}. \tag{19}$$

The general formula gives

$$c_{N,\epsilon}^{S^1} = \frac{1}{\frac{N}{2\pi} (4\pi)^{1/2} \epsilon^{3/2}} = \boxed{\frac{\sqrt{\pi}}{N \epsilon^{3/2}}.} \tag{20}$$

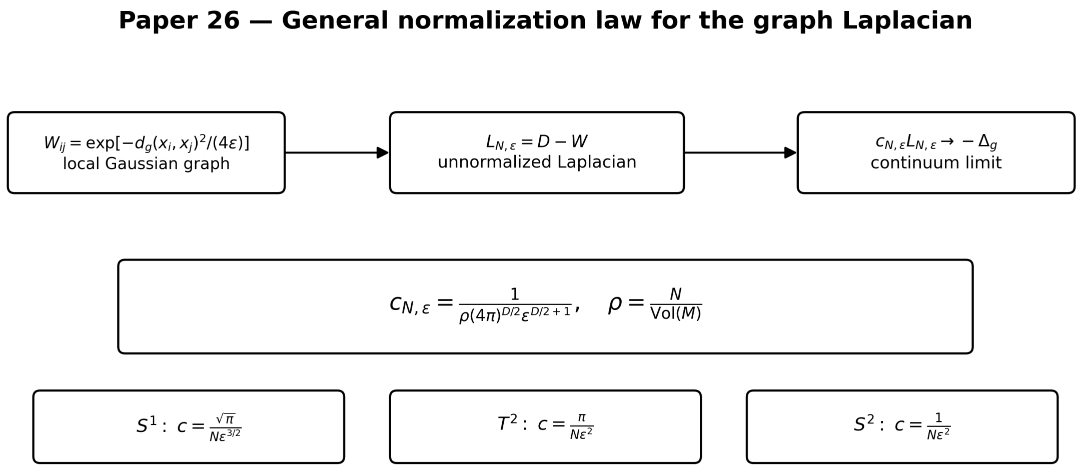

Figure 2: General normalization law for the unnormalized local Gaussian graph Laplacian. The normalization depends on the node density, the dimension and the kernel scale.

The continuum spectrum of $-\mathrm{d}^2/\mathrm{d}\theta^2$ is

$$0, \quad 1, 1, \quad 4, 4, \quad 9, 9, \ldots. \tag{21}$$

The numerical experiment estimates the normalization from the first nonzero spectral band and compares the normalized low spectrum with k^2.

The best prefactor result is

$$c_{N,\epsilon} N \epsilon^{3/2} = 1.7746703414, \tag{22}$$

to be compared with

$$\sqrt{\pi} = 1.7724538509. \tag{23}$$

The relative prefactor error is

$$1.25 \times 10^{-3}. \tag{24}$$

The low-spectrum relative errors are

$$\text{mean} = 1.23\%, \qquad \text{max} = 2.94\%. \tag{25}$$

7 Validation on T^2

For the flat square torus $T^2 = [0, 2\pi)^2$,

$$\mathrm{Vol}(T^2) = 4\pi^2, \qquad D = 2, \qquad \rho = \frac{N}{4\pi^2}. \tag{26}$$

The normalization is therefore

$$c_{N,\epsilon}^{T^2} = \frac{1}{\frac{N}{4\pi^2}(4\pi)\epsilon^2} = \boxed{\frac{\pi}{N\epsilon^2}.} \tag{27}$$

The continuum spectrum of the positive Laplacian on T^2 is

$$\lambda_{m,n} = m^2 + n^2. \tag{28}$$

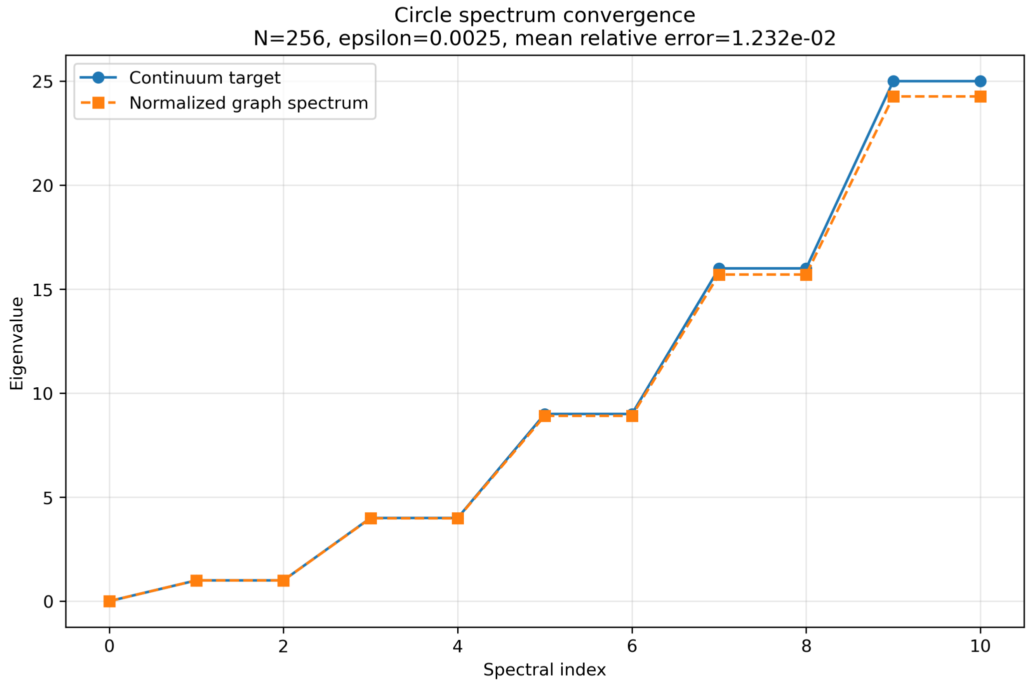

Figure 3: Low-spectrum convergence on the unit circle. After normalization by $\sqrt{\pi}/(N\epsilon^{3/2})$, the graph spectrum agrees with the target spectrum k^2 at the percent level.

Because a regular periodic grid is block-circulant, the spectrum can be computed efficiently by FFT. The corrected T^2 experiment gives

$$c_{N,\epsilon}N\epsilon^2 = 3.1573267968, \tag{29}$$

to be compared with

$$\pi = 3.1415926536. \tag{30}$$

The relative prefactor error is

$$0.5008\%. \tag{31}$$

The low-spectrum relative errors with the theoretical normalization are

$$\text{mean} = 2.46\%, \qquad \text{max} = 4.84\%. \tag{32}$$

8 Validation on S^2

For the unit sphere,

$$\mathrm{Vol}(S^2) = 4\pi, \qquad D = 2, \qquad \rho = \frac{N}{4\pi}. \tag{33}$$

The general law gives

$$c_{N,\epsilon}^{S^2} = \frac{1}{\frac{N}{4\pi}(4\pi)\epsilon^2} = \boxed{\frac{1}{N\epsilon^2}}. \tag{34}$$

The continuum spectrum is

$$\lambda_\ell = \ell(\ell+1), \qquad \text{multiplicity } 2\ell+1. \tag{35}$$

The tested low spectrum is therefore

$$0, \quad 2,2,2, \quad 6,6,6,6,6, \quad 12,\ldots. \tag{36}$$

Using a Fibonacci quasi-uniform sampling of the unit sphere, the best row gives

$$N = 1024, \qquad \epsilon = 0.01, \tag{37}$$

with empirical prefactor

$$c_{N,\epsilon} N \epsilon^2 = 1.0201663725. \tag{38}$$

The prefactor error is

$$2.02\%. \tag{39}$$

The low-spectrum relative errors are

$$\text{mean} = 6.85\%, \qquad \max = 10.35\%. \tag{40}$$

These errors are larger than in the S^1 and T^2 tests because the Fibonacci sphere is quasi-uniform but not an exact spectral grid, the graph is not circulant, and the spherical eigenvalues have nontrivial multiplicities.

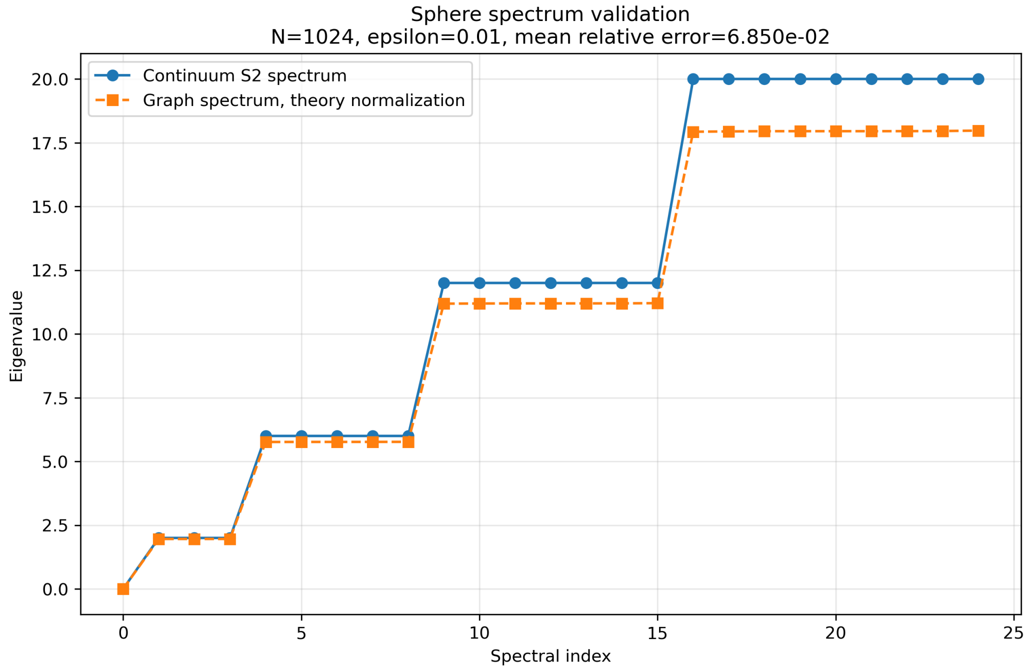

Figure 4: Low-spectrum validation on the unit sphere. The result confirms the predicted normalization $1/(N\epsilon^2)$, with larger finite-N and sampling errors than in the flat periodic cases.

9 Summary of the three uniform tests

The three controlled tests validate the same general normalization law:

$$c_{N,\epsilon} = \frac{1}{\rho (4\pi)^{D/2} \epsilon^{D/2+1}}. \tag{41}$$

10 Non-uniform density and drift bias

The preceding result assumes locally uniform density. This assumption is essential.

Geometry	Density ρ	Predicted $c_{N,\epsilon}$	Status
S^1	$N/(2\pi)$	$\sqrt{\pi}/(N\epsilon^{3/2})$	0.125% prefactor error
T^2	$N/(4\pi^2)$	$\pi/(N\epsilon^2)$	0.501% prefactor error
S^2	$N/(4\pi)$	$1/(N\epsilon^2)$	2.02% prefactor error

Table 1: Uniform-density validation of the general normalization law. The sphere is less sharp because of quasi-uniform sampling and curvature corrections, but still confirms the predicted prefactor.

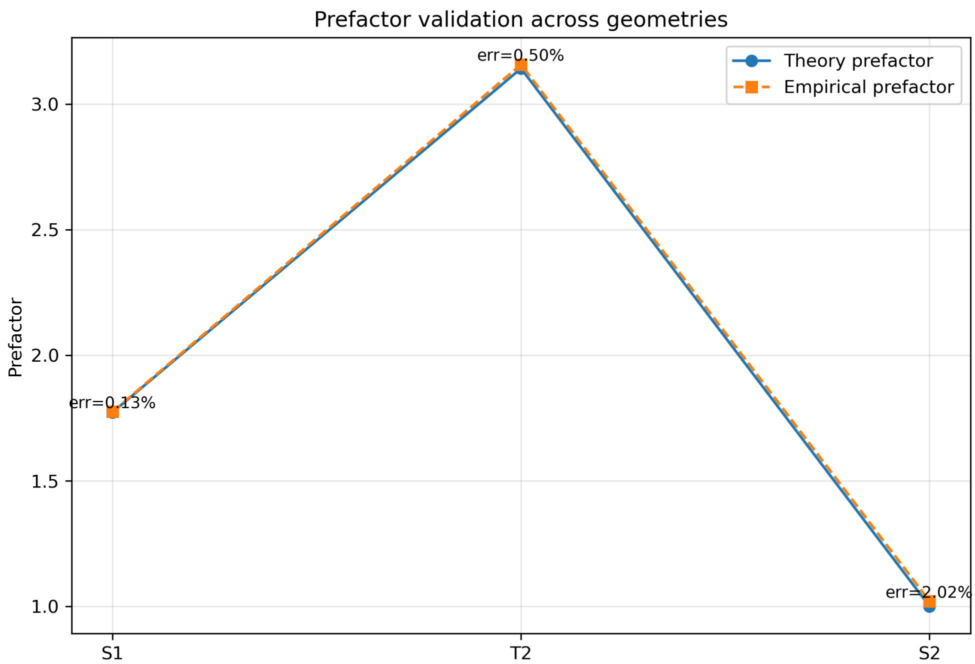

Figure 5: Prefactor validation across S^1, T^2, and S^2. The empirical prefactors agree with the theoretical values derived from the general density normalization law.

Let the density be position-dependent:

$$\rho = \rho(x). \tag{42}$$

Then

$$L_{N,\epsilon} f(x) \simeq \rho(x) \int k_\epsilon(x,y)\, [f(x) - f(y)]\, \mathrm{d}y. \tag{43}$$

Expanding both f and ρ, the raw combinatorial Laplacian contains a density drift. At leading order one obtains

$$L_{N,\epsilon} f \simeq -\rho(x)(4\pi)^{D/2} \epsilon^{D/2+1} \left[\Delta_g f + 2\nabla \log \rho \cdot \nabla f\right]. \tag{44}$$

After local normalization,

$$\boxed{c_{N,\epsilon}(x) L_{N,\epsilon} f \simeq -\Delta_g f - 2\nabla \log \rho \cdot \nabla f.} \tag{45}$$

The exact coefficient depends on the graph normalization convention, but the existence of a density-induced drift term is structural.

10.1 Controlled S^1 density test

We tested the non-uniform density

$$\rho(\theta) = \frac{1}{2\pi}(1 + a\cos\theta), \qquad a = 0.5, \tag{46}$$

with test function

$$f(\theta) = \sin(2\theta). \tag{47}$$

The drift term is controlled analytically:

$$\partial_\theta \log \rho = \frac{-a\sin\theta}{1 + a\cos\theta}. \tag{48}$$

The raw graph operator was compared to two targets:

$$\text{pure target} = -f''(\theta), \tag{49}$$
$$\text{drift target} = -f''(\theta) - 2(\partial_\theta \log \rho) f'(\theta). \tag{50}$$

The best run gives

$$N = 2048, \qquad \epsilon = 0.02, \tag{51}$$

with density contrast

$$\rho_{\max}/\rho_{\min} \simeq 3. \tag{52}$$

The raw operator errors are

$$\mathrm{RMSE}_{\text{pure}} = 0.3921, \qquad \mathrm{RMSE}_{\text{drift}} = 0.1738. \tag{53}$$

Thus the drift target improves the fit by a factor

$$2.26. \tag{54}$$

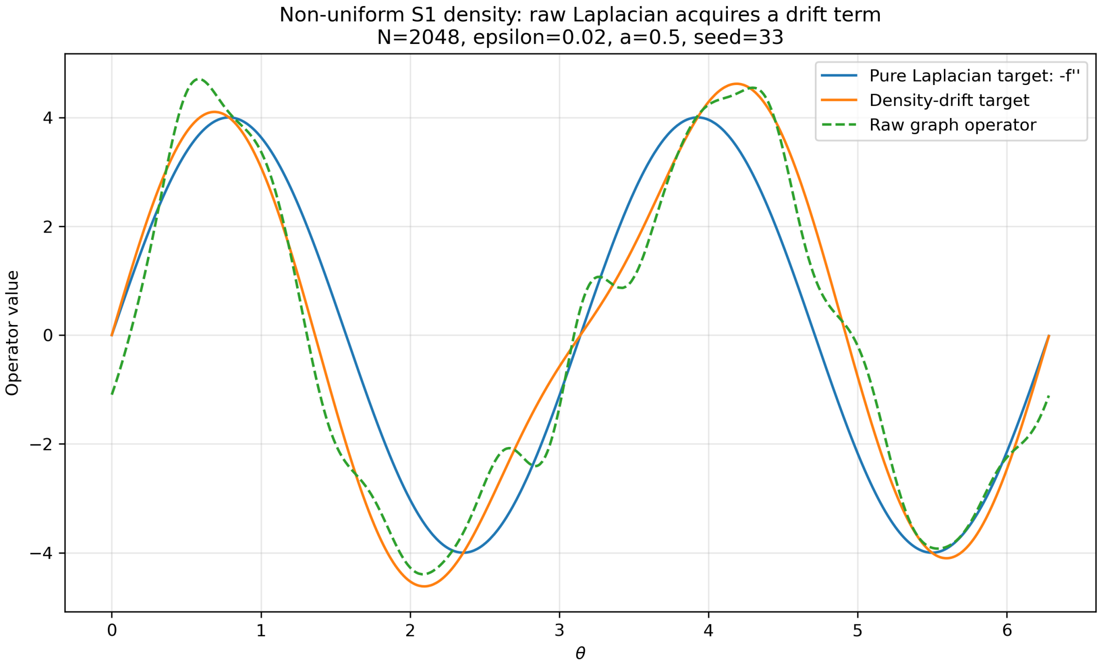

Figure 6: Controlled non-uniform S^1 density test. The raw graph Laplacian agrees better with the density-drift operator than with the pure Laplacian target.

11 Density-corrected Laplacians: two conventions

The density-bias test shows that the raw combinatorial Laplacian should not be used blindly in non-uniform sampling regimes. However, density correction is convention-dependent. Two graph-operator conventions must be distinguished:

1. a symmetric combinatorial correction,

$$L^{(\alpha)}_{\text{comb}} = D^{(\alpha)} - W^{(\alpha)};$$

2. the diffusion-maps Markov generator,

$$G^{(\alpha)} = \frac{I - P^{(\alpha)}}{\epsilon}.$$

This distinction is essential. A finite-N correction that performs best for the symmetric combinatorial operator is not necessarily the same correction that is optimal for the Markov diffusion-maps generator.

11.1 Symmetric combinatorial correction

Define

$$q_i = \sum_j W_{ij}. \tag{55}$$

We first tested the corrected symmetric kernel

$$W^{(\alpha)}_{ij} = \frac{W_{ij}}{q_i^\alpha q_j^\alpha}. \tag{56}$$

The associated combinatorial Laplacian is

$$L^{(\alpha)}_{\text{comb}} = D^{(\alpha)} - W^{(\alpha)}. \tag{57}$$

The tested cases were

$$\alpha = 0, \qquad \alpha = \frac{1}{2}, \qquad \alpha = 1. \tag{58}$$

The case $\alpha = 0$ is the raw graph. The case $\alpha = 1/2$ is the symmetric degree-flattening correction

$$W_{ij}^{(1/2)} = \frac{W_{ij}}{\sqrt{q_i q_j}}. \tag{59}$$

The median results are:

α	Pure RMSE	Drift RMSE	Degree contrast
0	0.4588	0.3803	3.3045
0.5	0.2690	0.2916	1.0622
1	0.3931	0.5163	3.0197

Table 2: Density-corrected non-uniform S^1 test for the symmetric combinatorial operator $D^{(\alpha)} - W^{(\alpha)}$. In this finite-$N$ convention, $\alpha = 1/2$ gives the best empirical recovery of the pure Laplacian and nearly flattens the degree distribution.

The best pure Laplacian recovery in this symmetric combinatorial family is

$$\alpha = \frac{1}{2}, \qquad \epsilon = 0.04, \qquad \mathrm{RMSE}_{\mathrm{pure}} = 0.16098. \tag{60}$$

The degree contrast is reduced to

$$1.0537. \tag{61}$$

This result should not be interpreted as a contradiction of diffusion-maps theory. The operator tested here is the symmetric combinatorial Laplacian $D^{(\alpha)} - W^{(\alpha)}$, not the row-normalized Markov generator used in the standard Coifman–Lafon construction.

11.2 Diffusion-maps Markov generator

To test the standard diffusion-maps convention, define

$$K_{ij}^{(\alpha)} = \frac{W_{ij}}{q_i^\alpha q_j^\alpha}. \tag{62}$$

The row-normalized Markov matrix is

$$P_{ij}^{(\alpha)} = \frac{K_{ij}^{(\alpha)}}{\sum_j K_{ij}^{(\alpha)}}. \tag{63}$$

The corresponding generator is

$$G^{(\alpha)} = \frac{I - P^{(\alpha)}}{\epsilon}. \tag{64}$$

This is the convention in which $\alpha = 1$ is expected to remove sampling density bias and recover the pure Laplace–Beltrami operator.

On the same non-uniform S^1 test, the Markov-generator results are:

The best pure Laplacian recovery is

$$\alpha = 1, \qquad \epsilon = 0.08, \qquad \mathrm{RMSE}_{\mathrm{pure}} = 0.0441. \tag{65}$$

The same row has

$$\mathrm{RMSE}_{\mathrm{drift}} = 0.3161. \tag{66}$$

Thus the diffusion-maps Markov generator resolves the apparent anomaly: $\alpha = 1/2$ is favoured only in the finite-N symmetric combinatorial construction, whereas $\alpha = 1$ is favoured in the row-normalized diffusion-maps convention.

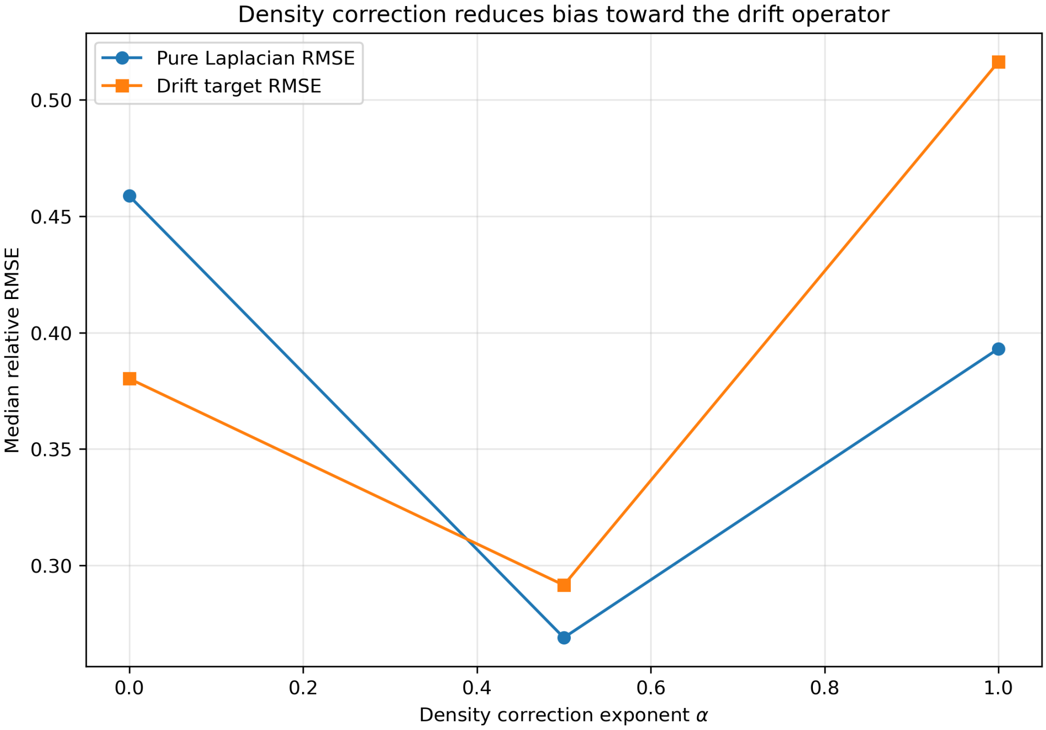

Figure 7: Density correction for the symmetric combinatorial operator. In this finite-N test, $\alpha = 1/2$ reduces the density bias and nearly flattens the graph degree distribution.

α	Median pure RMSE	Median drift RMSE
0	0.3757	0.2521
0.5	0.2343	0.2858
1	0.1216	0.3680

Table 3: Diffusion-maps Markov-generator test on the same non-uniform S^1 density. In the Markov convention $G^{(\alpha)} = (I - P^{(\alpha)})/\epsilon$, the expected choice $\alpha = 1$ gives the best recovery of the pure Laplacian.

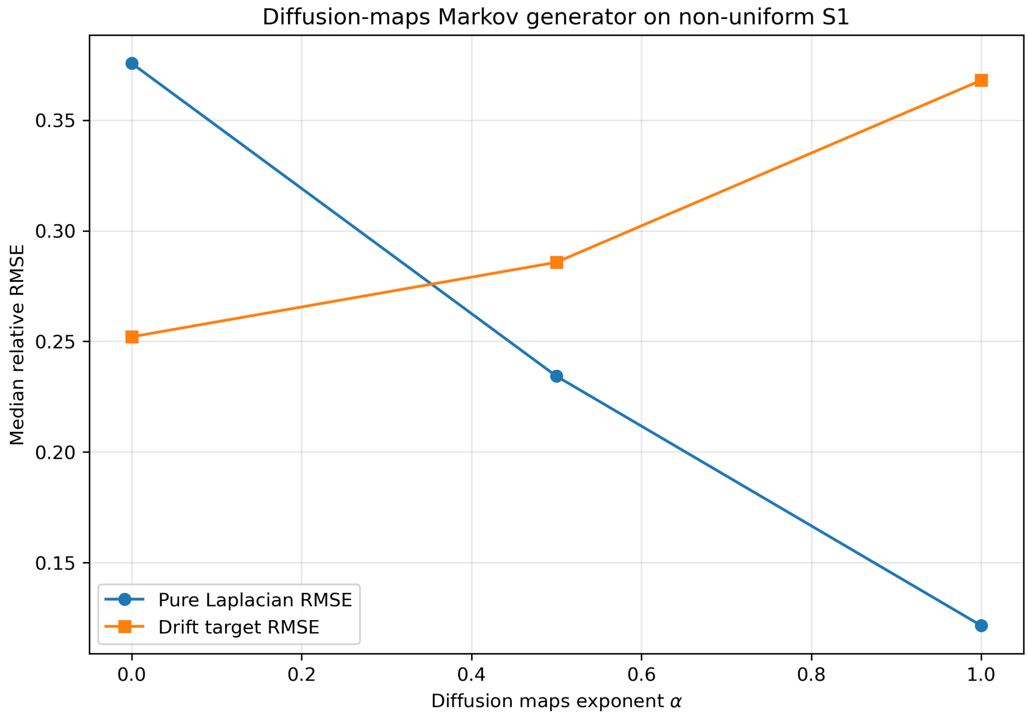

Figure 8: Diffusion-maps Markov-generator comparison. In the row-normalized Markov convention, $\alpha = 1$ gives the smallest median RMSE against the pure Laplace–Beltrami target.

12 Relation to mutual-information graphs

The numerical and analytical tests in this paper use local Gaussian kernels. BuP, however, starts from mutual information:

$$W_{ij} = I(i : j). \tag{67}$$

Therefore, the bridge from this paper to full BuP requires a local-kernel hypothesis:

$$I(i : j) \simeq F(d_g(x_i, x_j)), \tag{68}$$

with a local regime in which F admits a diffusion-kernel approximation.

A schematic example is

$$I(i : j) \sim \exp\left[-\frac{d_g(x_i, x_j)^2}{4\epsilon_{\rm ent}}\right] \tag{69}$$

at short entanglement distance.

This paper does not prove that arbitrary mutual-information graphs satisfy this assumption. Instead, it establishes the normalization and density correction required once the local-kernel regime is present.

13 Implications for SPARC and SLACS graphs

The density issue is not merely technical. Several BuP graphs used in the phenomenological programme are not uniformly sampled. In particular:

- SPARC-inspired graphs may be baryon-weighted;
- SLACS-inspired graphs may be lensing-weighted;
- micro-closure graphs may contain nontrivial entanglement-density profiles.

Thus one should not apply the raw result

$$c_{N,\epsilon} L_{N,\epsilon} \to -\Delta_g \tag{70}$$

blindly to those graphs.

The correct statement is:

> For locally uniform entanglement density, the raw graph Laplacian converges to the Laplace–Beltrami operator after the normalization
>
> $$c_{N,\epsilon} = \frac{1}{\rho(4\pi)^{D/2} \epsilon^{D/2+1}}.$$
>
> For non-uniform entanglement density, the raw Laplacian contains density drift, and a density-corrected graph Laplacian is required to recover a pure geometric operator.

This distinction strengthens rather than weakens the programme: it explains why different BuP sectors require explicit normalization choices.

14 Limitations

The results of Paper 26 should be read with the following limitations.

Local Gaussian assumption. The analytical derivation assumes a local Gaussian kernel. Full mutual information graphs require an additional physical argument establishing a local diffusion-kernel regime.

Uniform density. The clean convergence to $-\Delta_g$ holds under locally uniform density. Non-uniform graphs require density correction or explicit drift terms.

Finite N effects. The numerical tests are finite. The sphere test is especially sensitive to sampling and spectral multiplicities.

Operator convergence. The numerical evidence is primarily spectral and test-function based. A full operator convergence theorem remains a mathematical target.

Normalization convention. Different graph Laplacians — combinatorial, random-walk, symmetric normalized and diffusion-maps Markov generators — have different density-drift structures. The normalization must be stated together with the operator convention. In particular, the finite-N preference for $\alpha = 1/2$ in the symmetric combinatorial construction should not be confused with the standard diffusion-maps Markov result, where $\alpha = 1$ gives the best recovery of the pure Laplace–Beltrami operator in our controlled test.

15 Conclusion

Paper 26 establishes the normalization of the local unnormalized graph Laplacian required for convergence to the Laplace–Beltrami operator.

For a compact D-dimensional manifold with locally uniform density,

$$\rho = \frac{N}{\mathrm{Vol}(M)}, \tag{71}$$

the result is

$$\boxed{L_{N,\epsilon} f \simeq -\rho(4\pi)^{D/2} \epsilon^{D/2+1} \Delta_g f.} \tag{72}$$

Therefore,

$$\boxed{c_{N,\epsilon} = \frac{1}{\rho(4\pi)^{D/2} \epsilon^{D/2+1}}} \tag{73}$$

gives

$$\boxed{c_{N,\epsilon} L_{N,\epsilon} \to -\Delta_g.} \tag{74}$$

The formula is validated on S^1, T^2, and S^2, giving the concrete normalizations

$$S^1 : \quad \frac{\sqrt{\pi}}{N\epsilon^{3/2}}, \qquad T^2 : \quad \frac{\pi}{N\epsilon^2}, \qquad S^2 : \quad \frac{1}{N\epsilon^2}. \tag{75}$$

The non-uniform S^1 tests show that the raw combinatorial Laplacian develops a density drift. In the symmetric combinatorial family $D^{(\alpha)} - W^{(\alpha)}$, the correction

$$W_{ij} \mapsto \frac{W_{ij}}{\sqrt{q_i q_j}}$$

substantially improves recovery of the pure Laplace–Beltrami operator in the finite-N controlled setting. In the diffusion-maps Markov-generator convention

$$G^{(\alpha)} = \frac{I - P^{(\alpha)}}{\epsilon},$$

the expected density-correcting choice $\alpha = 1$ gives the best recovery, with a best pure-Laplacian RMSE of 0.0441.

Thus Paper 26 closes the principal normalization debt identified in Paper 25 and clarifies the density-correction required before applying the entanglement Laplacian to non-uniform BuP graphs.

References

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